Data C100/200 - MIDTERM 3

Spring 2026

Write your name BIG and clearly: ______________________________

Email: ______________________________@berkeley.edu

Student ID: ______________________________

Name and SID of the person on your left: ______________________________

Name and SID of the person on your right: ______________________________

Exam Room:______________________ Seat Number: ______________________

Instructions:

This exam consists of **16 points** spread out over **6 questions**. The exam must be completed in **50 minutes** unless you have accommodations supported by a DSP letter.

- Note that some questions have circular bubbles to select a choice. This means that you should only **select one choice**. Other questions have boxes. This means you should **select all that apply**. Please **shade in** the circle/box **fully** to mark your answer.
- Blank answers and incorrect answers are graded identically, so it's in your best interest to answer every question.
- **You MUST write your Student ID number at the top of each page.**
- You should not use a calculator, scratch paper, or notes you own other than the reference sheets distributed at the beginning of the exam.

Honor Code:

As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. I am the person whose name is on the exam, and I completed this exam in accordance with the Honor Code.

Signature: ______________________________

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1 Every Midterm Has a SQL [4 Pts]

For each query below, determine the size of the SQL output. Specifically, choose the number of **rows** and the number of **columns** in the final result table.

(a) [2 Pts] Consider table `daphne` and the following SQL query.

`daphne`

x	s	z	w
12	6	2	39
32	8	9	37
20	11	32	48
20	18	38	1
29	14	43	22

```
SELECT d.x, MIN(d.z)
FROM daphne d
WHERE d.x < 27
GROUP BY d.x
HAVING MIN(d.s) > 7
```

(i) How many rows are in the output?

○ 0 ○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6

(ii) How many columns are in the output?

○ 0 ○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6

(b) [2 Pts] Consider tables `josh` and `ramesh` and the following SQL query.

`josh`

y	b
15	6
9	16
13	27

`ramesh`

x	a
22	14
26	13
7	9

```
SELECT j.b, r.a
FROM josh j
LEFT JOIN ramesh r
ON j.y = r.a
```

(i) How many rows are in the output?

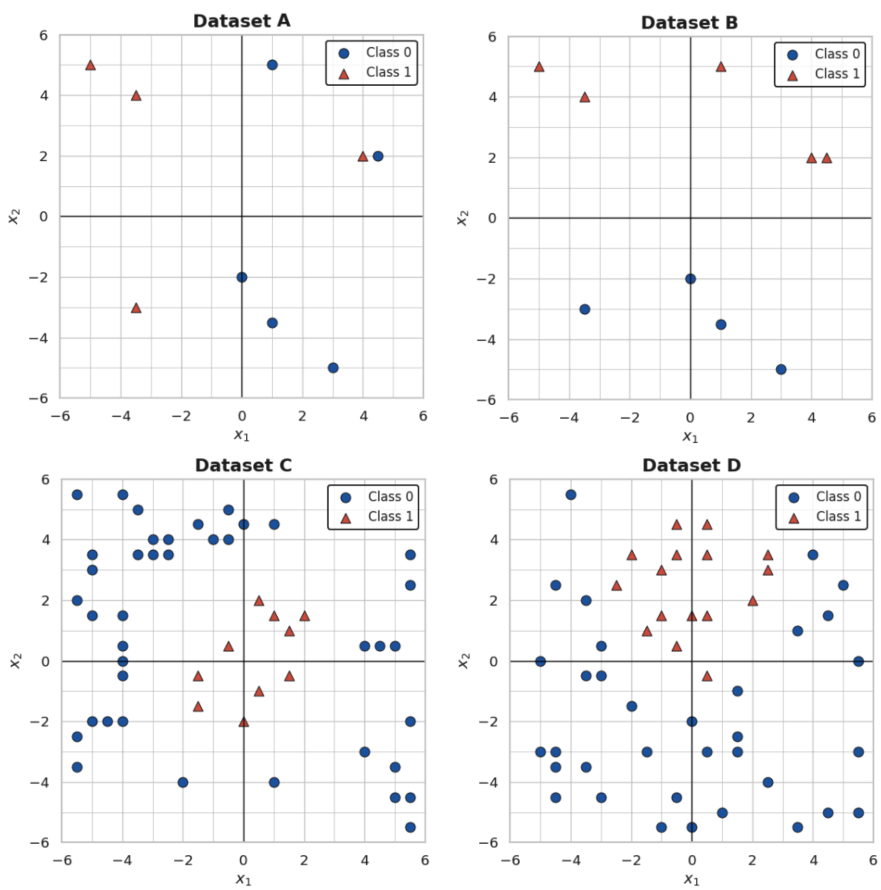
○ 0 ○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6 ○ 7 ○ 8 ○ 9

(ii) How many columns are in the output?

○ 0 ○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6 ○ 7 ○ 8 ○ 9

2 Data 100 Exams Never Have Regressive Logistics [5 Pts]

The four scatter plots below show four different datasets (A, B, C, and D). Each dataset has two features x_1 and x_2 and binary class labels y, where ◯ denotes Class 0 and △ denotes Class 1. In the plots, x_1 refers to the **horizontal** axis and x_2 refers to the **vertical** axis. *All points in the datasets are fully visible in the plots: None are hidden behind the legend, and there is no overplotting.*



(a) [2 Pts] For each dataset, evaluate this statement: **Logistic regression with features x_1 and x_2 (and no intercept) can achieve perfect classification accuracy on this dataset.** *Assume the model fitting process always converges.*

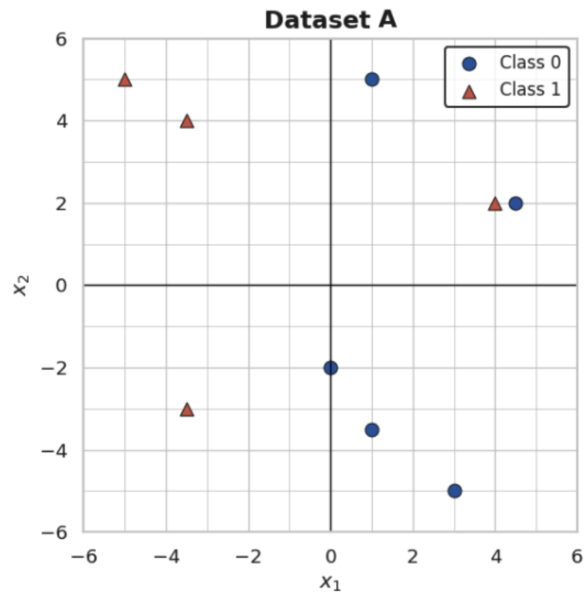
Dataset A ◯ True ◯ False ◯ Not enough information to answer

Dataset B ◯ True ◯ False ◯ Not enough information to answer

Dataset C ◯ True ◯ False ◯ Not enough information to answer

Dataset D ◯ True ◯ False ◯ Not enough information to answer

(b) [2 Pts] Consider Dataset A. Suppose we use a classifier that predicts Class 1 (△) for any point where $x_1 > 1.5$, and Class 0 (◯) otherwise. Compute precision and recall for this classifier on Dataset A. *For your convenience, Dataset A is printed again below:*



Compute **recall** in the box below. Draw a box around your final answer.

Compute **precision** in the box below. Draw a box around your final answer.

(c) [1 Pt] *This part does not depend on any of the plots or datasets above.* The probability of Event A is $2/5$. Event B has double the odds of Event A. What are the odds of Event B, **and** what is the probability of Event B?

If there is not enough information to determine the oods or probability, write ***NA***.

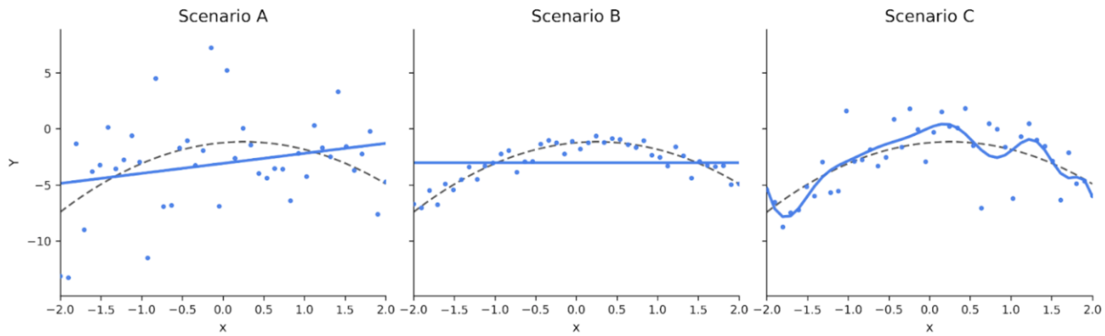
Show your work in the boxes below. Draw a box around your final answer. You can leave your final answer as an unsimplified algebraic expression (e.g., $5(2 + 3) - 1$).

Compute the **odds** of Event B in this box:

Compute the **probability** of Event B in this box:

3 We Found Bias in a Noisy Place! [3 Pts]

Consider the following 3 plots, each visualizing a different scenario. Each scenario involves a different data-generating process (dashed line and points), and an OLS regression fit as either a constant model, SLR model, or higher-order polynomial model (solid line). The DGPs are very similar to the one described in Lecture 19: You can find a detailed description at the end of this question, but you shouldn't need it for your answers.



(a) [1 Pt] Rank the **irreducible error** of the scenarios.

Highest irreducible error: ◯ Scenario A ◯ Scenario B ◯ Scenario C

Lowest irreducible error: ◯ Scenario A ◯ Scenario B ◯ Scenario C

(b) [1 Pt] Rank the **model bias** of the scenarios.

Highest model bias: ◯ Scenario A ◯ Scenario B ◯ Scenario C

Lowest model bias: ◯ Scenario A ◯ Scenario B ◯ Scenario C

(c) [1 Pt] Rank the **model variance** of the scenarios.

Highest model variance: ◯ Scenario A ◯ Scenario B ◯ Scenario C

Lowest model variance ◯ Scenario A ◯ Scenario B ◯ Scenario C

Details about the DGPs and OLS models

Remember, you might not need all this information for your answers: we're providing it to rule out edge cases and make sure the question is clear.

Each plot's DGP has the following form: $y_i = f(x_i) + \epsilon_i$, where ϵ_i is a random variable with mean 0 and constant variance V_J. i indicates the observation number and J indicates the plot label. The set of x_i's in each plot are identical. Each of the three DGPs has a unique value of V. In other words, $V_A \neq V_B \neq V_C$.

f is a deterministic polynomial function. The dashed line in each plot shows f, which is the same for all three DGPs. The solid line in each plot is an OLS model fit with specification $\hat{F}_J$, where J indicates the plot label. So, there are three OLS model specifications in total: $\hat{F}_A$, $\hat{F}_B$, and $\hat{F}_C$. Each specification $\hat{F}_J$ can be a constant model, linear model, or polynomial of any degree. All models were fit with no regularization.

Assume that the observed data points are representative of their respective DGP. In other words, none of the observed data points are unusual outliers.

Nothing you write on this page will be graded. The exam continues on the next page.

4 This Question Is Not Very PCA [1.5 Pts]

Suppose you have a dataset with **573 observations** and **4 linearly independent features**. You run PCA on this dataset.

The table below shows the variance of each feature, ordered from greatest to least:

Feature	**Variance**
Feature 1	60
Feature 2	51
Feature 3	21
Feature 4	19

(a) [0.5 Pts] Suppose you share all of the latent feature vectors with a colleague. How many **principal component vectors** must you share with your colleague so that they can **fully reconstruct** the original dataset? If there is not enough information to answer the question, write **NA**. *Assume the original data was not centered, standardized, or transformed in any other way before running PCA.*

(b) [0.5 Pts] The variance of **Feature 1 (variance = 60)** __________ the variance of **PC1**.

Hint: It is always possible for the two variances to be equal.

- ○ must be **greater** than or equal to
- ○ must be **equal** to
- ○ must be **less** than or equal to
- ○ `(Cannot be determined)`

(c) [0.5 Pts] The variance of **Feature 2 (variance = 51)** __________ the variance of **PC2**.

Hint: It is always possible for the two variances to be equal.

- ○ must be **greater** than or equal to
- ○ must be **equal** to
- ○ must be **less** than or equal to
- ○ `(Cannot be determined)`

5 Data 100 Cereal: Now with Clusters! [2 Pts]

(a) [1 Pt] Consider the following one-dimensional dataset: **2, 6, 10, 15**

Suppose you run a modified version of K-medians clustering on these four data points, where **K = 2**.

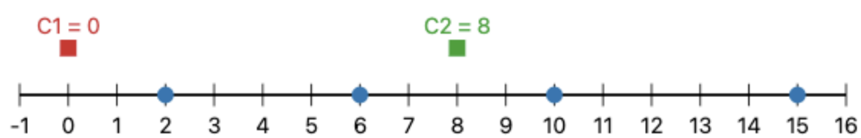
In one dimension, K-Medians clustering is exactly the same as K-means clustering, but **it uses the median instead of the mean to define cluster centers**.

Initial position of the cluster centers:

- Cluster center 1: **0**
- Cluster center 2: **8**

After a **single iteration** of K-medians clustering, where are the cluster centers located?

Note: This question is not asking you to report the cluster centers after the K-medians algorithm has fully converged. Just run a single iteration. If you need to find the median of an even number of points, use the average of the two middle points.



Cluster Center 1: | |

Cluster Center 2: | |

(b) [1 Pt] Consider these two clusters of a **new** one-dimensional dataset: $\{3, 1, 2\}$ and $\{7, 9, 8\}$

What is the **single linkage** distance between these two clusters?

What is the **complete linkage** distance between these two clusters?

6 Fishy Coefficients [1.5 Pts]

Consider a DataFrame called `df` with 100 rows. Here are the first two rows of `df`:

	numEvents	temp	humidity	uvIndex
0	0.7	80	84	8.8
1	0.7	55.7	45.9	1.8

Josh fits an OLS model that predicts `uvIndex` from the features `numEvents`, `temp`, and `humidity`. Josh includes an intercept in his model. Josh does not transform any feature.

For Josh's fitted model, $\hat{\theta}_{numEvents} = 6$ and the intercept is 5.

Enter **NA** if there is not enough information to answer a question.

(a) [0.5 pts] Suppose that for a particular observation, all feature values are 0. What is the predicted value of `uvIndex` for this observation?

(b) [0.5 pts] Suppose two observations have identical values for all features except `numEvents`, which is 4 units larger for one of the observations. What is the absolute difference between the *true* value of `uvIndex` for the first observation and the *true* value of `uvIndex` for the second observation?

(c) [0.5 pts] Consider a single observation. Suppose we increase the value of `numEvents` for that observation by 3 without changing the value of any other feature. What is the absolute difference between the *predicted* value of `uvIndex` before the change and the *predicted* value of `uvIndex` after the change?

Please state any relevant assumptions in the box below (Optional).

You are done with the midterm- Congratulations!

Regex Crossword (Optional, not graded)

Fill each square with a single CAPITAL letter so that every row and column matches its corresponding regular expression.

<table>
<tr><th></th><th>H[GIN\s]+</th><th>[UMN\s]*</th><th>\w\s(U|M|N)+</th><th>([GIN]|&|\s)+</th><th></th></tr>
<tr><td>\w*</td><td></td><td></td><td></td><td></td><td>(HU|GI)+</td></tr>
<tr><td>(\w)N\W&</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>\WM[M-Z]+</td><td></td><td></td><td></td><td></td><td>[^O-T]+U.</td></tr>
<tr><td>[GIN]+.</td><td></td><td></td><td></td><td></td><td>[H-K].+\s</td></tr>
</table>

"^" matches the position at the beginning of string (unless used for negation "[^]")

"$" matches the position at the end of string character.

"?" match preceding literal or sub-expression 0 or 1 times.

"+" match preceding literal or sub-expression *one* or more times.

"*" match preceding literal or sub-expression *zero* or more times

"." match any character except new line.

"[ ]" match any one of the characters inside, accepts a range, e.g., "[a-c]".

"( )" used to create a sub-expression

"\d" match any *digit* character. "\D" is the complement.

"\w" match any *word* character (letters, digits, underscore). "\W" is the complement.

"\s" match any *whitespace* character including tabs and newlines. \S is the complement.

"{m,n}" The preceding element or subexpression must occur between m and n times, inclusive.